URBANVERSE

MEMBER 3 | GIS DATA + MAP INTEGRATION

Project: City-Scale Digital Twin for Multi-Hazard Prediction and Urban Resilience Planning

Hackathon target: Build a convincing, working flood-risk MVP for a small pilot area in Mangalore. Use real map/geographical data where practical and clearly label simulated rainfall and risk outputs.

Your mission

Provide usable geographical data and help make the map represent a real pilot area.

Your tasks

- Agree with the team on one small pilot area in Mangalore. Do not collect data for the whole city.
- Find usable map/geographical sources for roads, water bodies, buildings and boundaries.
- Prepare a small GeoJSON file for the selected area, with stable zone IDs.
- If available, provide elevation/terrain data for the pilot area.
- Record each dataset's source, licence, date, units and limitations.
- Help Member 2 display the GeoJSON on the map.
- Check that the zone IDs in the GeoJSON match the simulation output.

What you must hand over

- data/pilot-area.geojson (or an agreed equivalent).
- At least 2–5 clearly identified zones for the MVP.
- docs/data-sources.md with links, licence notes and limitations.
- A short field guide describing each GeoJSON property and zone ID.
- A screenshot or short test showing the data displayed on the map.

File ownership and coordination

- Own data/ and docs/data-sources.md.
- Do not silently rename zone IDs after Member 4 starts using them.
- Do not claim a dataset is live or authoritative unless its source confirms that.
- If elevation data cannot be obtained quickly, report that and provide a simpler fallback.

Start in this order

1. Pick the small pilot boundary with the team.
2. Get a basic map layer and create stable zone IDs.
3. Share a tiny sample GeoJSON with Members 1, 2 and 4 early.
4. Help Member 2 render the zones.
5. Add extra roads, water and elevation layers only after the base map works.

Your acceptance test

Open the GeoJSON in the frontend and verify that every zone appears in the correct place and has the exact stable zone_id used by the simulation.

Team rule

Commit working changes to your own Git branch, push them, and tell the integration lead what changed. Do not edit another member's files without agreement. If you are blocked, share the exact error and the smallest reproducible example rather than silently changing the shared interface.